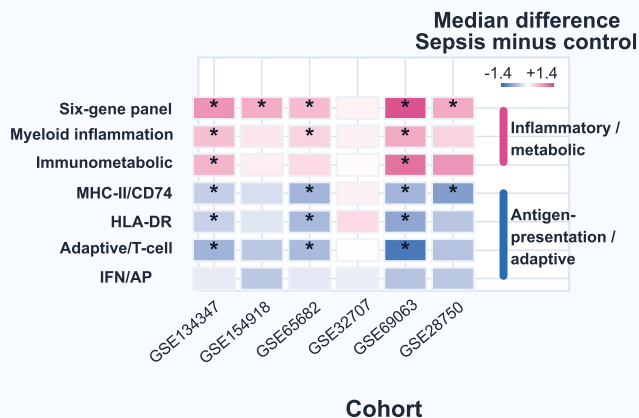


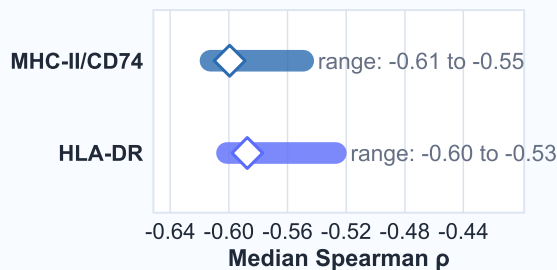
A Signature direction atlas

Control-available cohorts; GSE54514 excluded. * FDR<0.05.



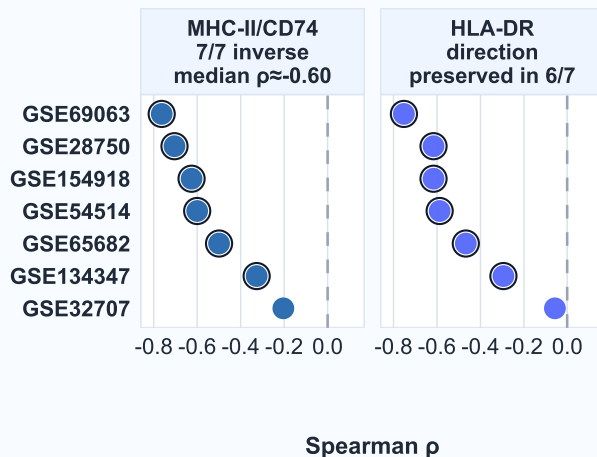
C Leave-one-cohort sensitivity

Direction preserved after sequential cohort removal.



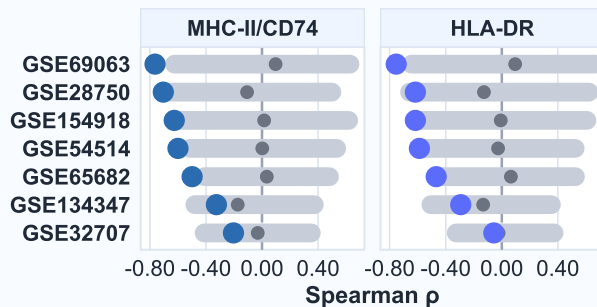
B Primary inverse coupling across cohorts

Observed ρ ; black outer ring or * indicates FDR<0.05.



D Random-signature null intervals

Observed six-gene ρ = color point.
Random 95% = grey bar; median = grey dot.



Gene-count matched random signatures;
not expression-level matched.